

# Classical NLP

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# 1- Tokenization

Tokenization is a fundamental step in Natural Language Processing (NLP) where a piece of text is split into smaller units called tokens. These tokens can be:

- Words (word-level tokenization)
- Characters (character-level tokenization)
- Subwords (e.g., using Byte Pair Encoding or WordPiece)

Example – Word Tokenization:

Input:

"I love NLP!"

Output tokens:

["I", "love", "NLP", "!"]

# 1- Tokenization

Tokenization helps machines:

- Understand the structure of text
- Prepare input for tasks like translation, classification, or question answering
- Handle different languages and punctuation properly

Types of Tokenization:

Word Tokenization

Sentence Tokenization

Subword Tokenization

Character Tokenization

## 2- Stop Words

Stop words are common words in a language that are often filtered out before or during natural language processing (NLP) tasks because they are considered to carry little meaningful information.

◆ Examples of English Stop Words:

- Articles: a, an, the
- Pronouns: I, you, he, they
- Prepositions: in, on, at, with
- Conjunctions: and, but, or
- Others: is, are, was, to, of, for



# 3-Stemming & Lemmatization

Stemming

Definition:

Process of reducing inflected words to their word stem.

Examples:

- history → histori
- historical → histori
- finally → fina
- final → fina
- finalized → fina

# 3-Stemming & Lemmatization

## Lemmatization

### Definition:

Process of reducing words to their base or dictionary form (lemma).

### Examples:

- history → history
- historical → history
- finally → final
- final → final
- finalized → final

# 4- Bag of words

Input Sentences:

- Sent1 → He is a good boy
- Sent2 → She is a good girl
- Sent3 → Boy & girl are good

Step 1: Remove Stop Words

- Sent1 → good boy
- Sent2 → good girl
- Sent3 → boy girl good



# 5- TF-IDF

Sentences:

- Sent1 → good boy
- Sent2 → good girl
- Sent3 → boy girl good

## 1. Term Frequency (TF):

Formula:

$$TF = \frac{\text{No. of repetitions of word in sentence}}{\text{Total words in the sentence}}$$

| Word | Sent1         | Sent2         | Sent3         |  |
|------|---------------|---------------|---------------|--|
| good | $\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{1}{3}$ |  |
| boy  | $\frac{1}{2}$ | 0             | $\frac{1}{3}$ |  |
| girl | 0             | $\frac{1}{2}$ | $\frac{1}{3}$ |  |



# 5- TF-IDF

Sentences:

- Sent1 → good boy
- Sent2 → good girl
- Sent3 → boy girl good

## 2. Inverse Document Frequency (IDF):

Formula:

$$\text{IDF} = \log \left( \frac{N}{n} \right)$$

Where:

- **N** = total number of sentences (documents)
- **n** = number of sentences containing the word

| Word | IDF             |
|------|-----------------|
| good | $\log(3/3) = 0$ |
| boy  | $\log(3/2)$     |
| girl | $\log(3/2)$     |

# 5- TF-IDF

Sentences:

- Sent1 → good boy
- Sent2 → good girl
- Sent3 → boy girl good

## 3. TF × IDF Output:

|       | $f_1$ (good) | $f_2$ (boy)                    | $f_3$ (girl)                   |  |
|-------|--------------|--------------------------------|--------------------------------|---------------------------------------------------------------------------------------|
| Sent1 | 0            | $\frac{1}{2} \times \log(3/2)$ | 0                              |                                                                                       |
| Sent2 | 0            | 0                              | $\frac{1}{2} \times \log(3/2)$ |                                                                                       |
| Sent3 | 0            | $\frac{1}{3} \times \log(3/2)$ | $\frac{1}{3} \times \log(3/2)$ |                                                                                       |